



Department of Biomedical Engineering

Digital Logic Design Lab

Lab Report:

Seven Segment Display

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Experiment:

Decimal to Seven Segment

Objectives:

- To design a **4-bit decimal-to-binary encoder** using OR gates.
- To design a **binary-to-7-segment decoder** using logic gates.
- To verify encoder and decoder truth tables through Proteus.
- To integrate both circuits to build a **decimal-to-display system**.
- To understand the difference between encoder and decoder operations.

Abstract- *In this experiment, we designed and simulated a decimal-to-binary encoder and a binary-to-seven-segment decoder using logic gates only. The encoder converts one active decimal input (0–9) into a 4-bit binary code, while the decoder converts the binary code into seven segment (a–g) outputs that display the corresponding digit on a 7segment display. Proteus simulation was used to verify the truth table, logic design, and the overall functionality. This experiment helped us understand how encoding and decoding processes work together in digital display systems.*

Apparatus:

- Proteus Design Suite
- Seven Segment Display (Common Cathode preferred)
- Logic gates (AND, OR, NOT, NAND, NOR)
- Push Buttons (for decimal input 0–9)
- Power supply (5V DC)

Introduction

Encoders and decoders are fundamental combinational logic circuits in digital electronics.

Encoder

An encoder, specifically a decimal-to-binary encoder, converts a single active input from many possible inputs into a corresponding binary code.

In this experiment, a 10-to-4 encoder is designed. The logic for each of the four binary output bits is created using a combination of OR gates, as each output bit is the logical OR of the selected decimal inputs that should result in that bit being '1'.

Decoder

A decoder, specifically a binary-to-seven-segment decoder, performs the inverse operation.

It accepts a 4-bit binary input and outputs signals to activate the necessary segments (labeled a through g) of a 7-segment display to visually represent the decimal digits 0 through 9.

Integrated Display System

In this system, the encoder's output is directly connected to the decoder's input, forming a complete signal path:

Decimal Input → Encoder → Binary → Decoder → 7-Segment Display

This integrated structure is crucial for applications such as **calculators, counters, and digital clock**

Task 1: Decimal to Binary Encoder:

Design 4-bit encoder that converts 10 input lines (0–9) into 4-bit binary output (D_0 , D_1 , D_2 and D_3).

Procedure:

The procedural method used to perform this experiment is given below:

1. Ten push buttons (0–9) were placed in Proteus, each representing one decimal input.
2. Pull-down resistors were used to ensure stable LOW when switches were unpressed.
3. OR gates (7432) were placed for generating output bits D_3 , D_2 , D_1 , D_0 .
4. Each binary output bit was wired according to the Proteus circuit design:
 - $D_0 = \text{OR}(\text{inputs } 8 \text{ and } 9)$
 - $D_1 = \text{OR}(\text{inputs } 4\text{--}7)$
 - $D_2 = \text{OR}(\text{inputs } 2, 3, 6, 7)$
 - $D_3 = \text{OR}(\text{inputs } 1, 3, 5, 7, 9)$
5. LEDs were connected to observe binary output.
6. Only one button was pressed at a time to verify the result.
7. The observed binary outputs were recorded in the truth table.

1. Truth Table Verification:

Decimal Input	Output			
Decimal	D_0	D_1	D_2	D_3
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0

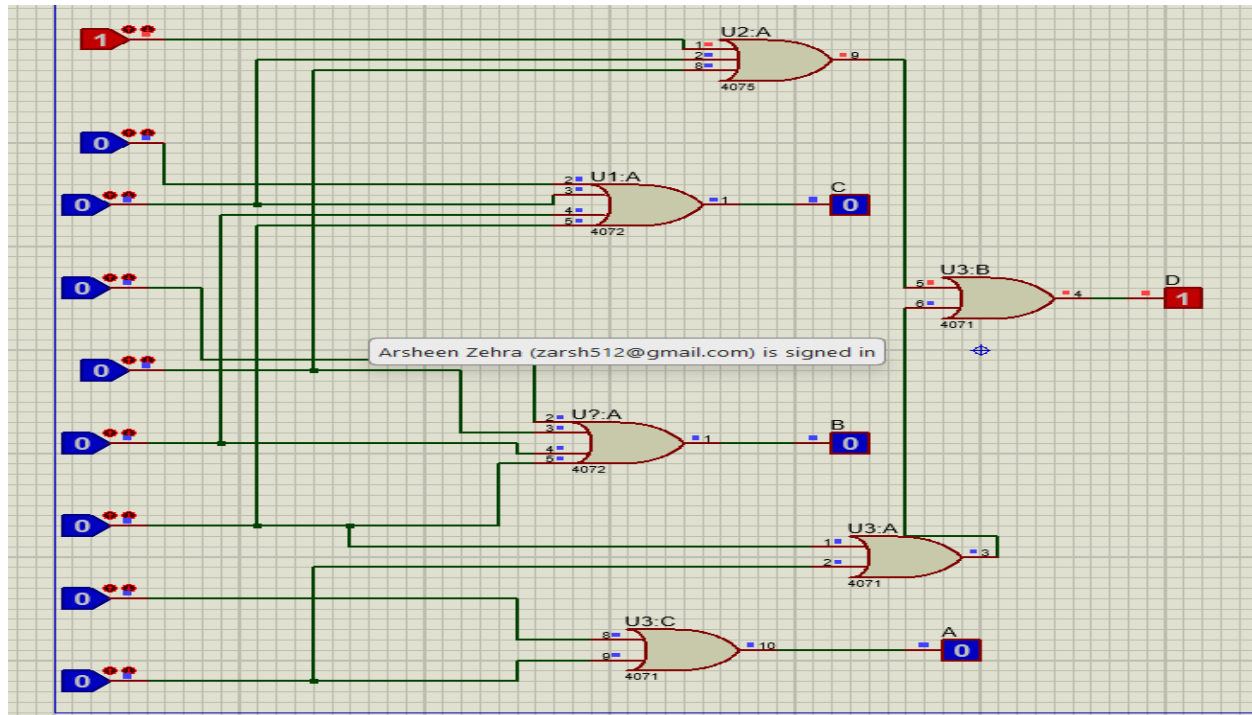
9	1	0	0	1
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2. Boolean Expression:

$$\begin{aligned}
 D_0 &= 8 + 9 \\
 D_1 &= 4 + 5 + 6 + 7 \\
 D_2 &= 2 + 3 + 6 + 7 \\
 D_3 &= 1 + 3 + 5 + 7 + 9
 \end{aligned}$$

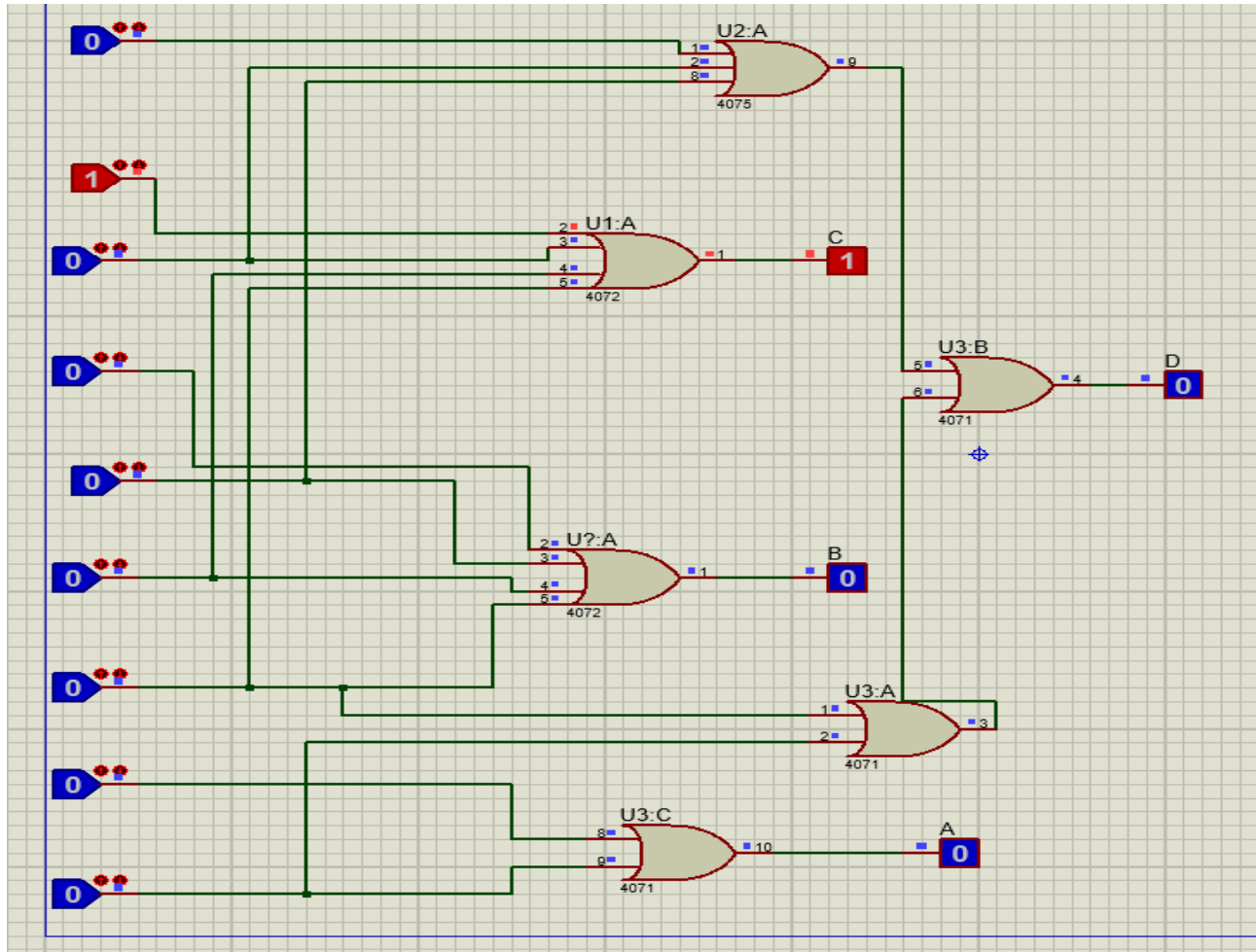
3. Proteus Simulation Screenshots:

1. Output at 1:



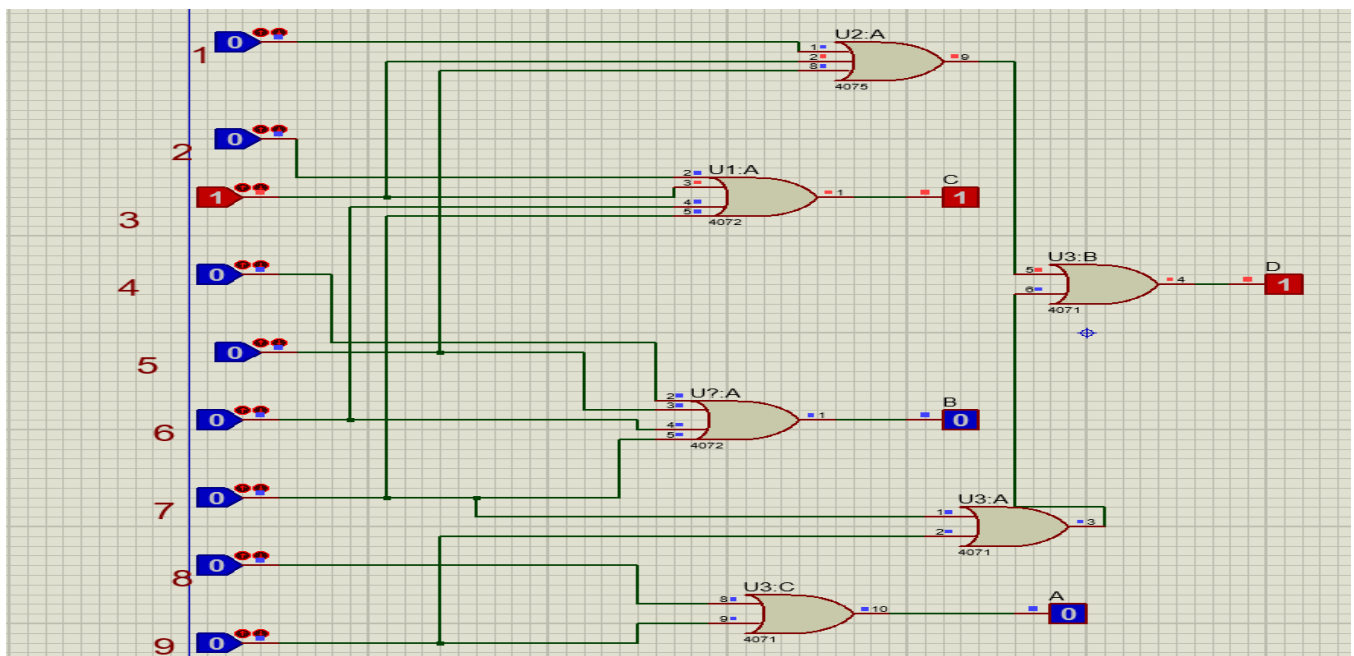
When input 1 will be HIGH output at D will also be HIGH.

2. Output at 2



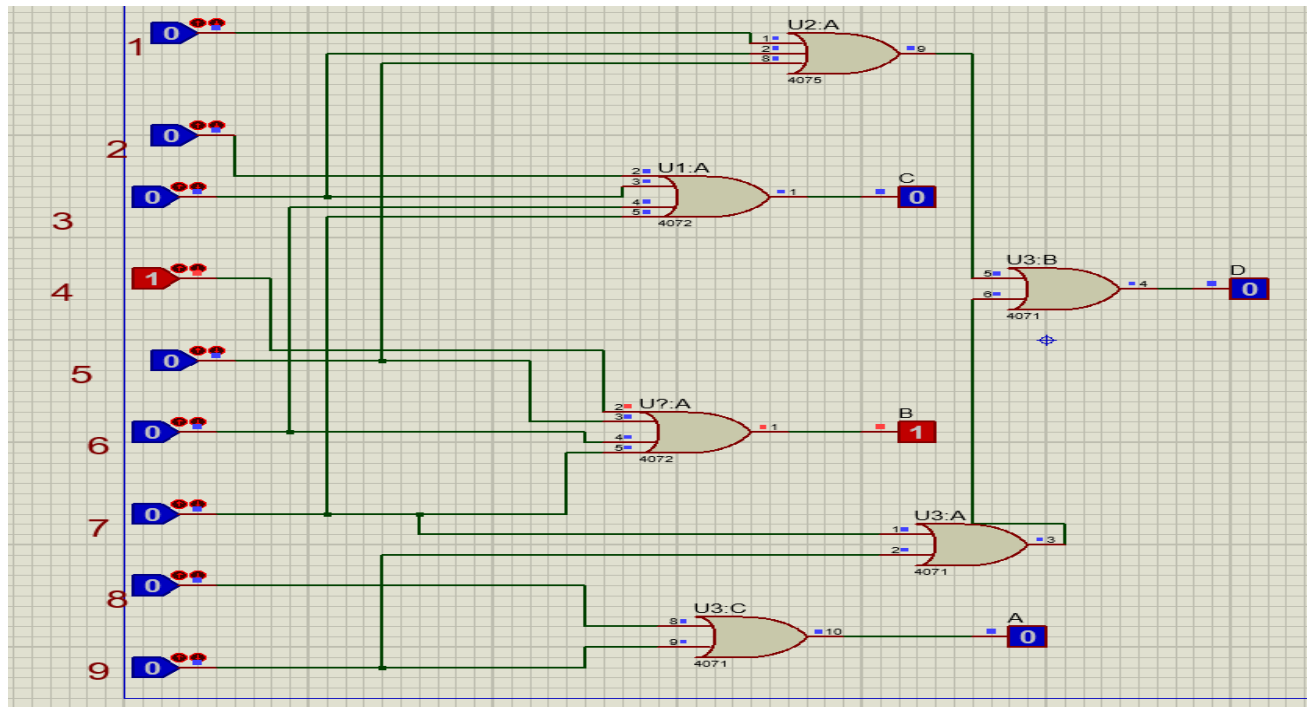
When input 2 will be HIGH output at C will also be HIGH

3. Output at 3:



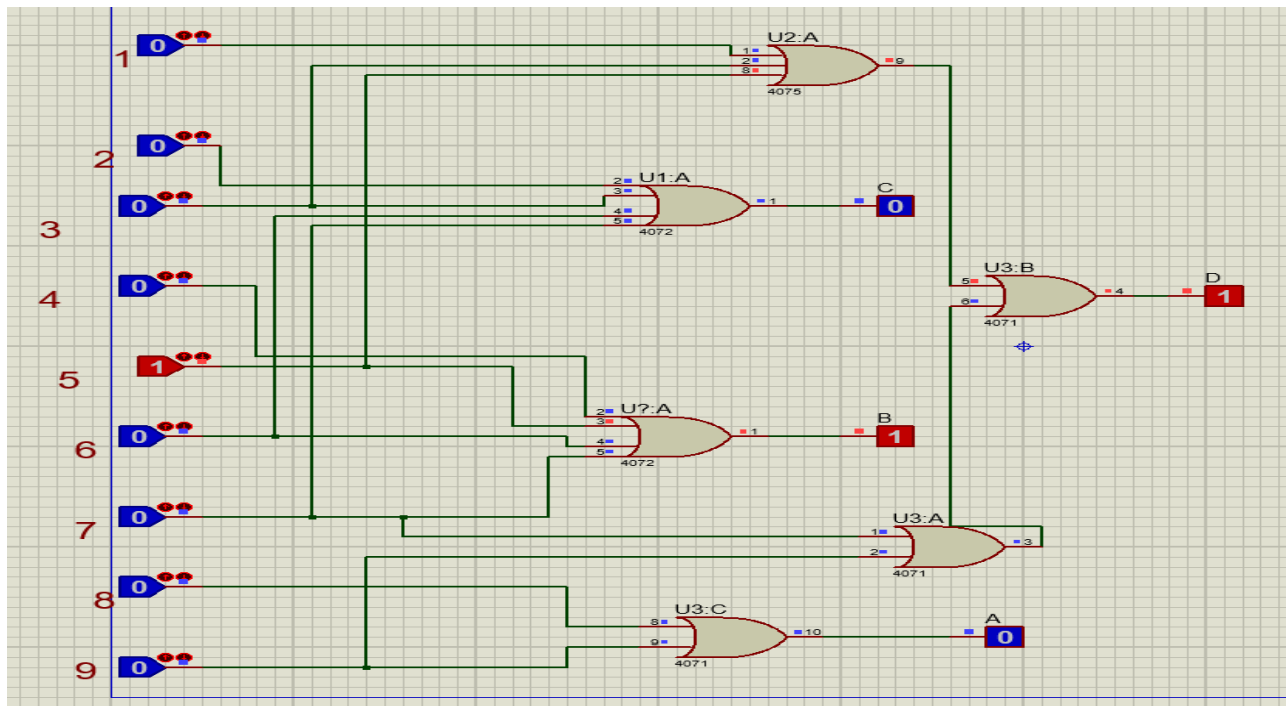
When input 3 will be HIGH output at D₂ and D₃ will also be HIGH.

4. Output at 4



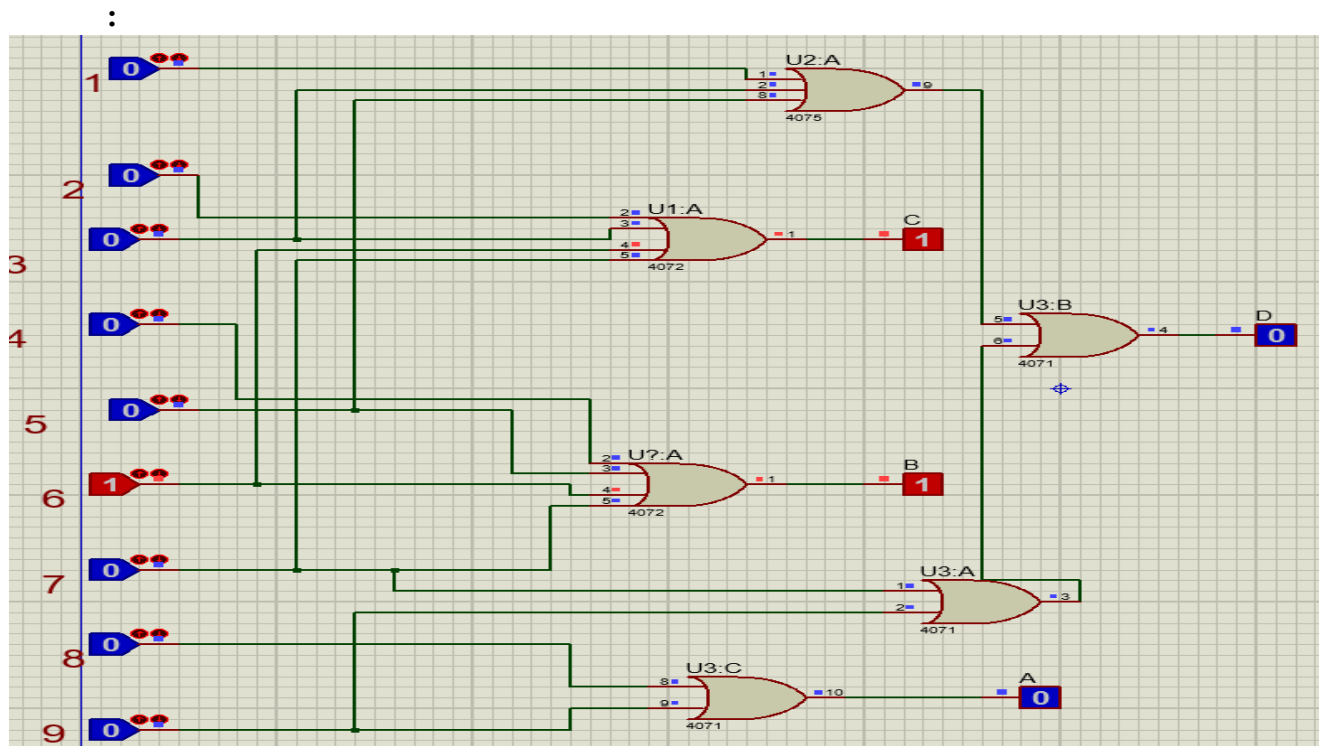
When input 4 will be HIGH output at D₁ will also be HIGH.

5. Output at 5:



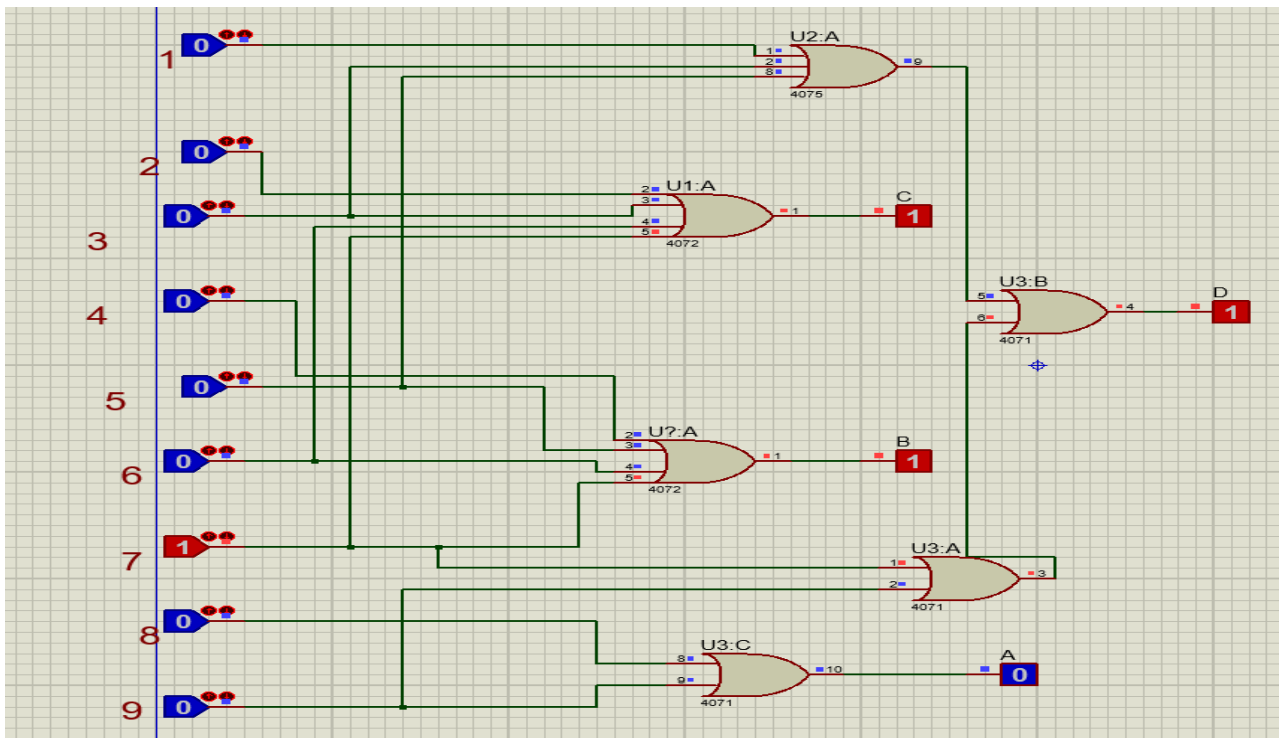
When input 4 will be HIGH output at D₁ and D₃ will also be HIGH.

6. Output at 6



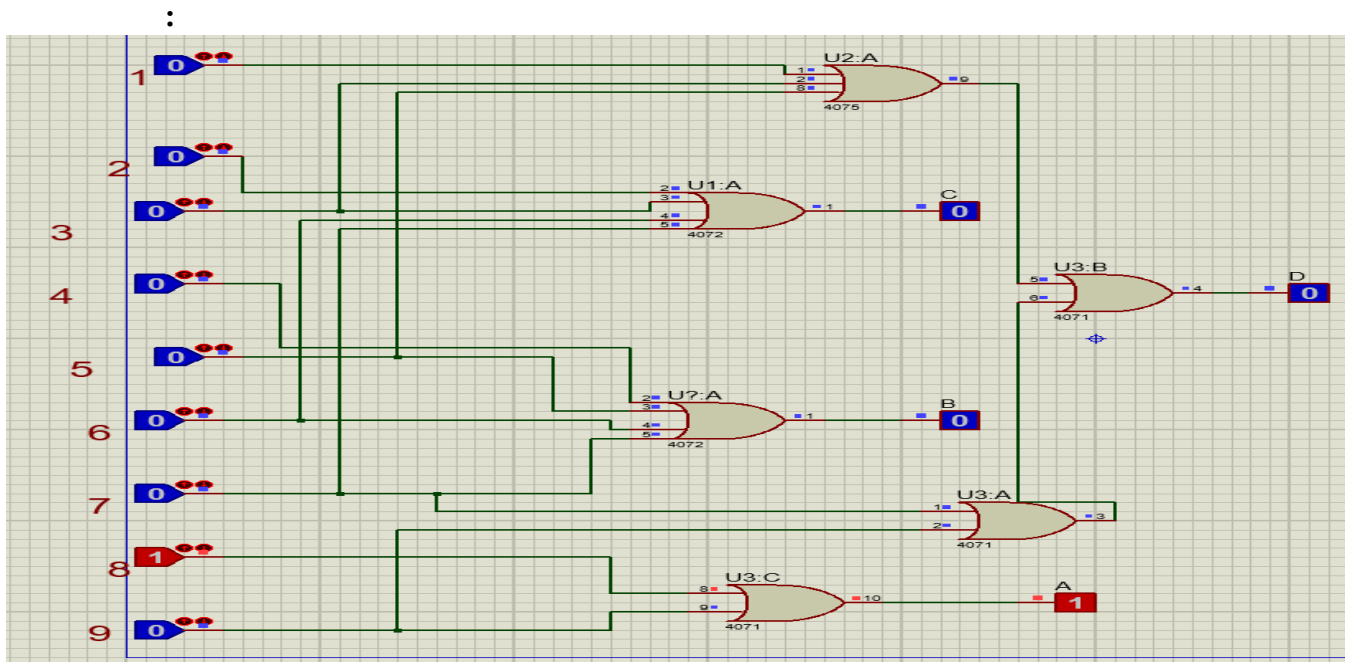
When input 6 will be HIGH output at D₁ and D₂ will also be HIGH.

7. Output at 7:



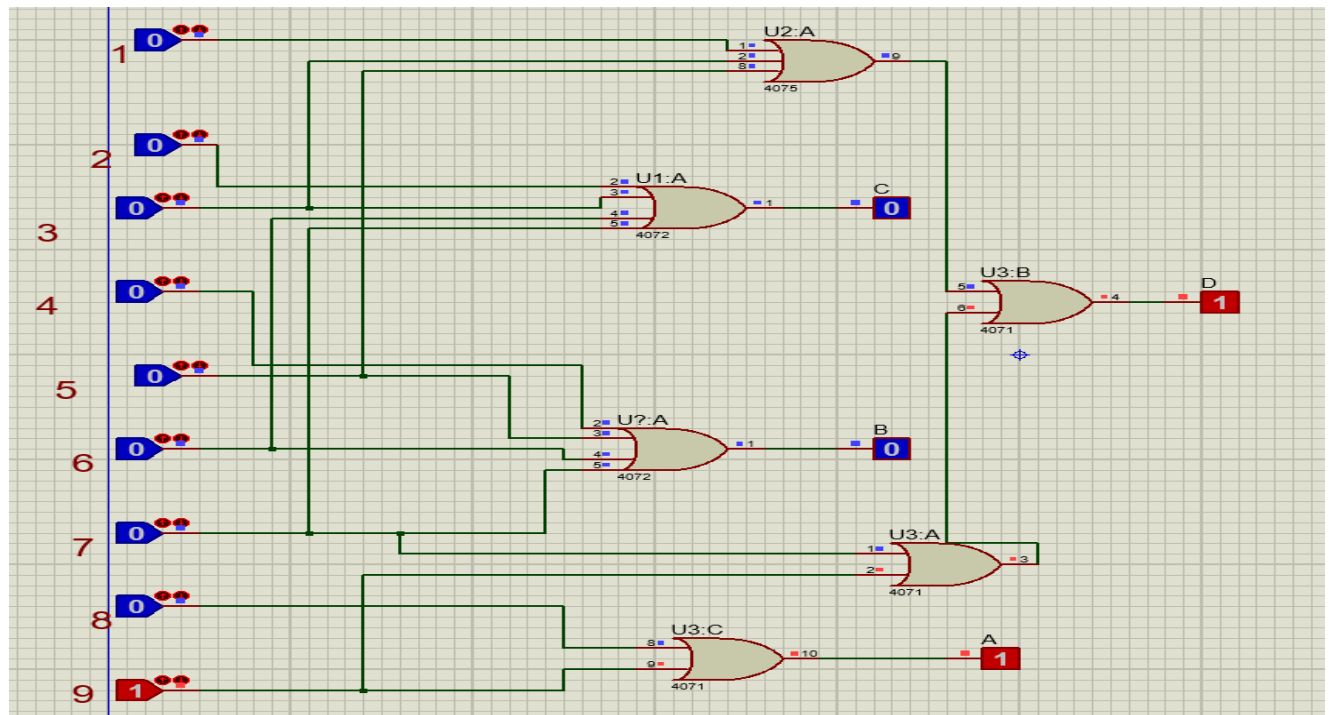
When input 7 will be HIGH output at D₁, D₂ and D₃ will also be HIGH.

8. Output at 8



When input 8 will be HIGH output at D₀ will also be HIGH.

9. Output at 9:



When input 6 will be HIGH output at D₀ and D₃ will also be HIGH.

At a time only one input is pressed and output is seen.

Task 2: Binary to Seven Segment Decoder:

Design a logic circuit that converts 4-bit binary input into seven segment control signals (a–g).

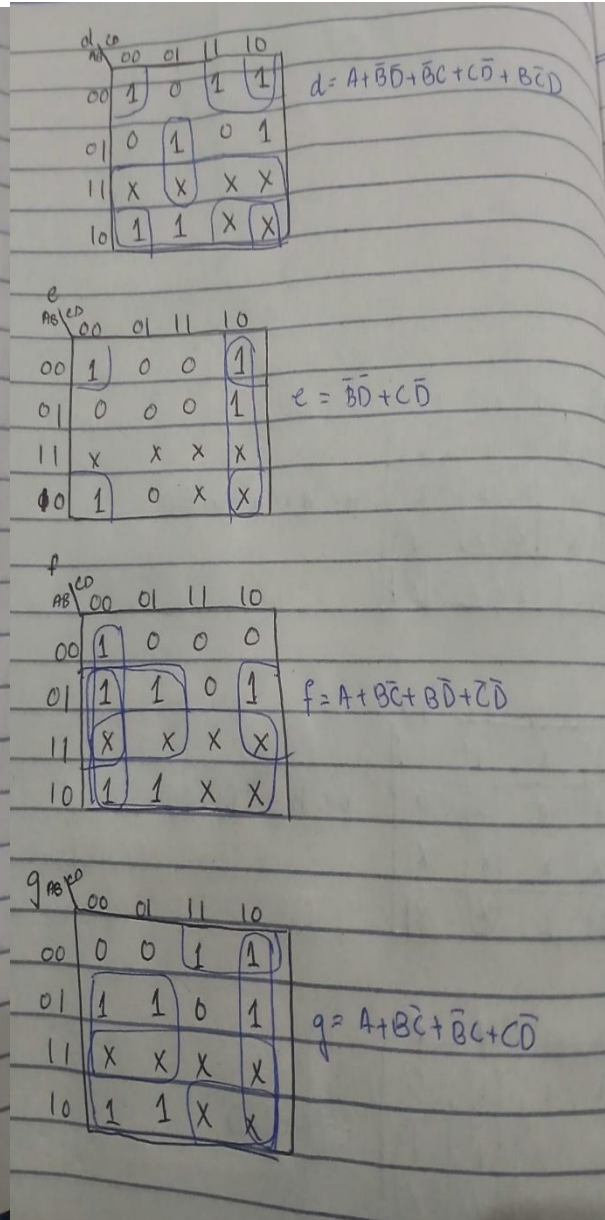
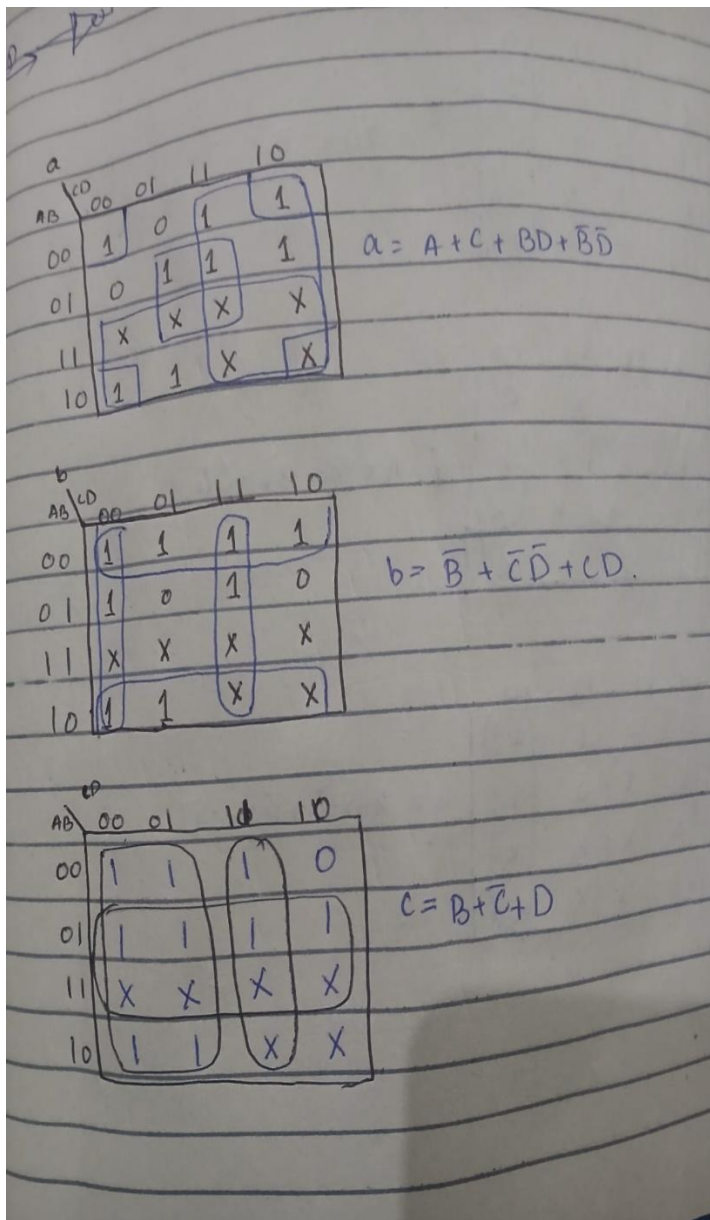
Procedure:

The procedural method used to perform this experiment is given below:

1. Seven AND/OR/NOT gate combinations were implemented to derive the Boolean expressions for segments a–g.
2. A 4-bit input (D₃ D₂ D₁ D₀) from Task 1 was connected.
3. For each decimal digit, segment conditions were verified.
4. Output of each logic circuit was connected to a segment LED of the 7-segment display.
5. All digits (0–9) were tested by applying corresponding binary input.

1. Truth Table Verification:

Input				Output						
A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	1	1	1	1	1	1	0
0	0	0	1	0	1	1	0	0	0	0
0	0	1	0	1	1	0	1	1	0	1
0	0	1	1	1	1	1	1	0	0	1
0	1	0	0	0	1	1	0	0	1	1
0	1	0	1	1	0	1	1	0	1	1
0	1	1	0	1	0	1	1	1	1	1
0	1	1	1	1	1	1	0	0	0	0
1	0	0	0	1	1	1	1	1	1	1
1	0	0	1	1	1	1	1	0	1	1



2. Boolean Expression:

$$a = A + C + B.D + B'.D'$$

$$b = B' + C.D + C'.D'$$

$$c = B + D + C'$$

$$d = A + C.B' + C.D' + B'.D' + B.D.C'$$

$$e = C.D' + B'.D'$$

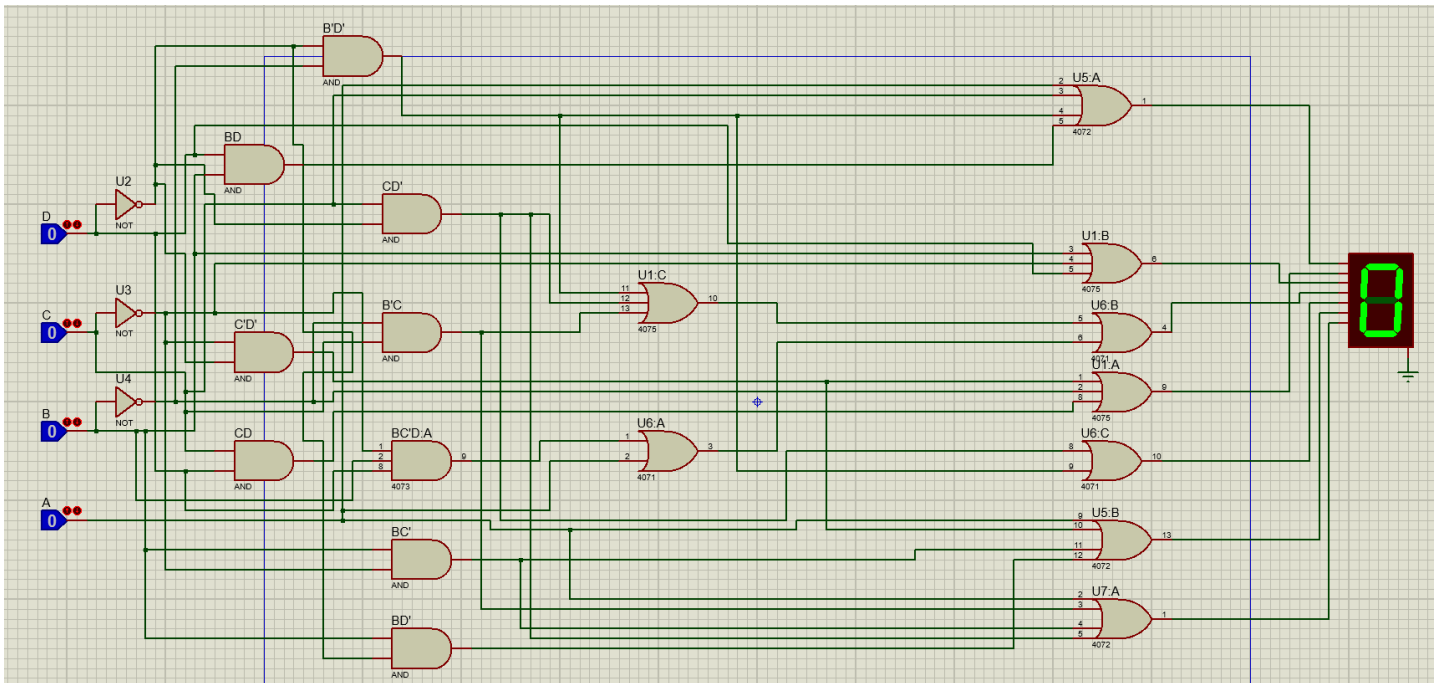
$$f = A + B.C' + C.B' + C.D'$$

$$g = A + B'.C + C.D' + B.C'$$

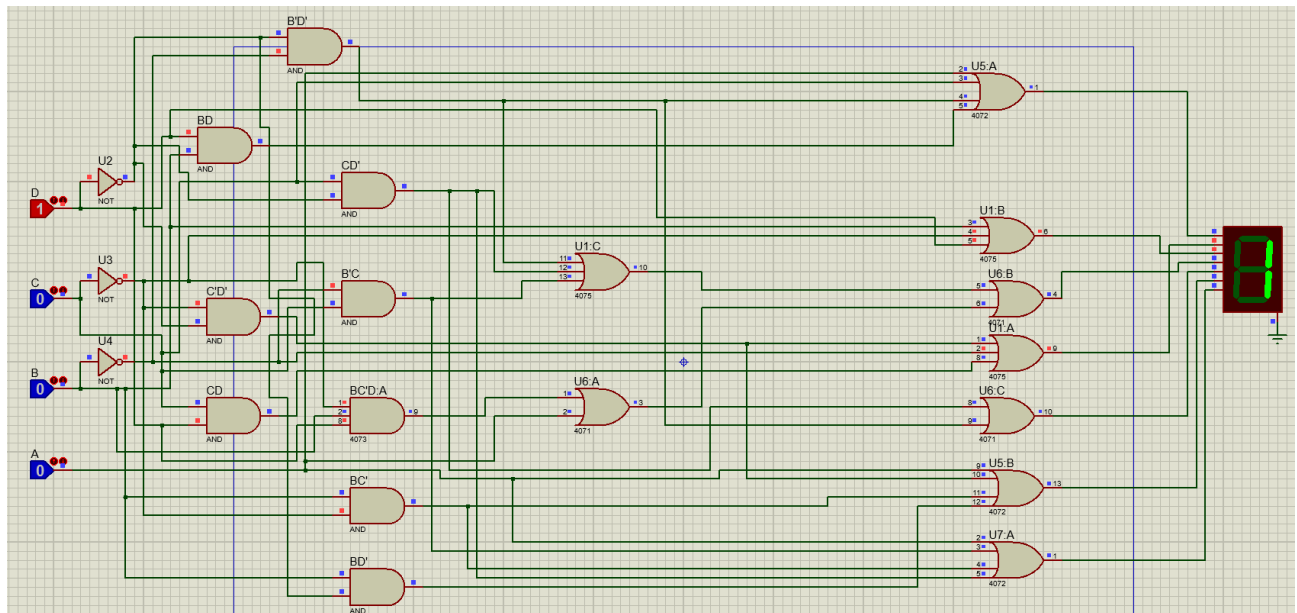
These are minimized expressions with inputs 0–9 as ON terms and 10–15 treated as don't-cares.

3. Proteus Simulation Screenshots:

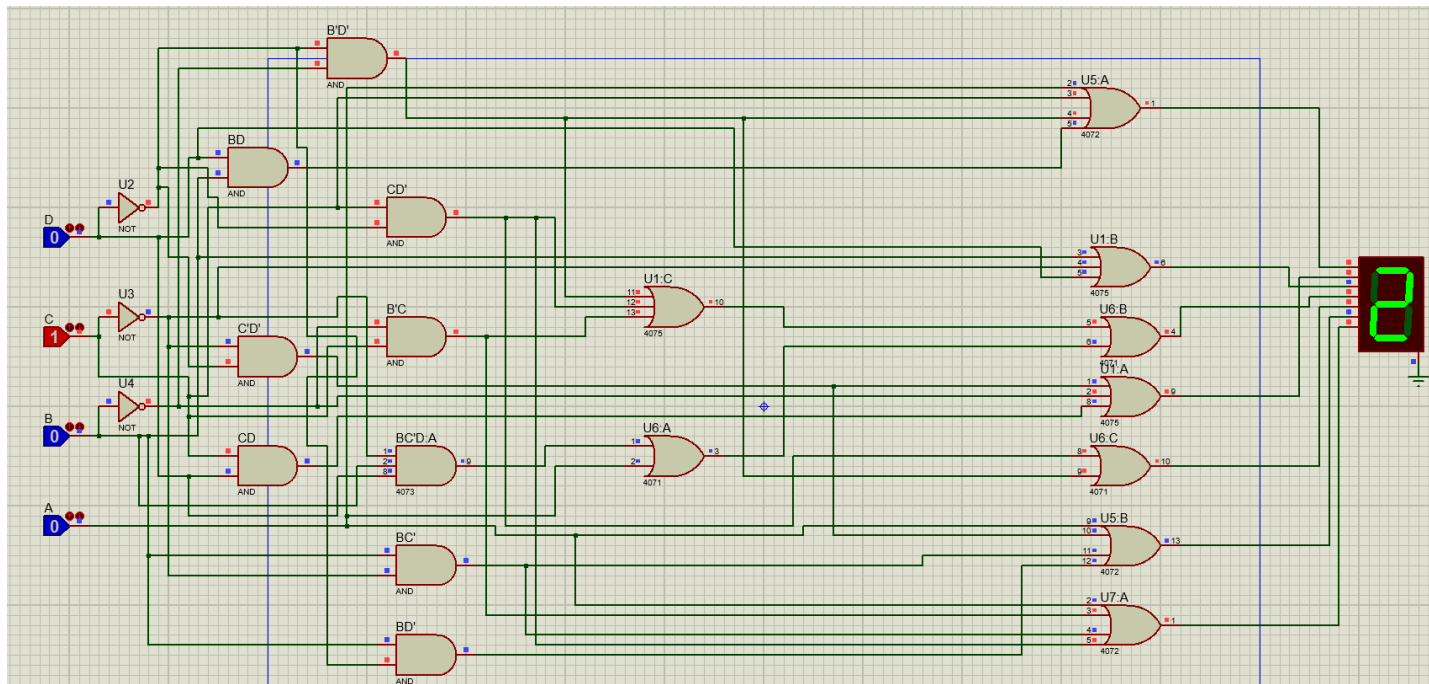
1. 0 is displayed:



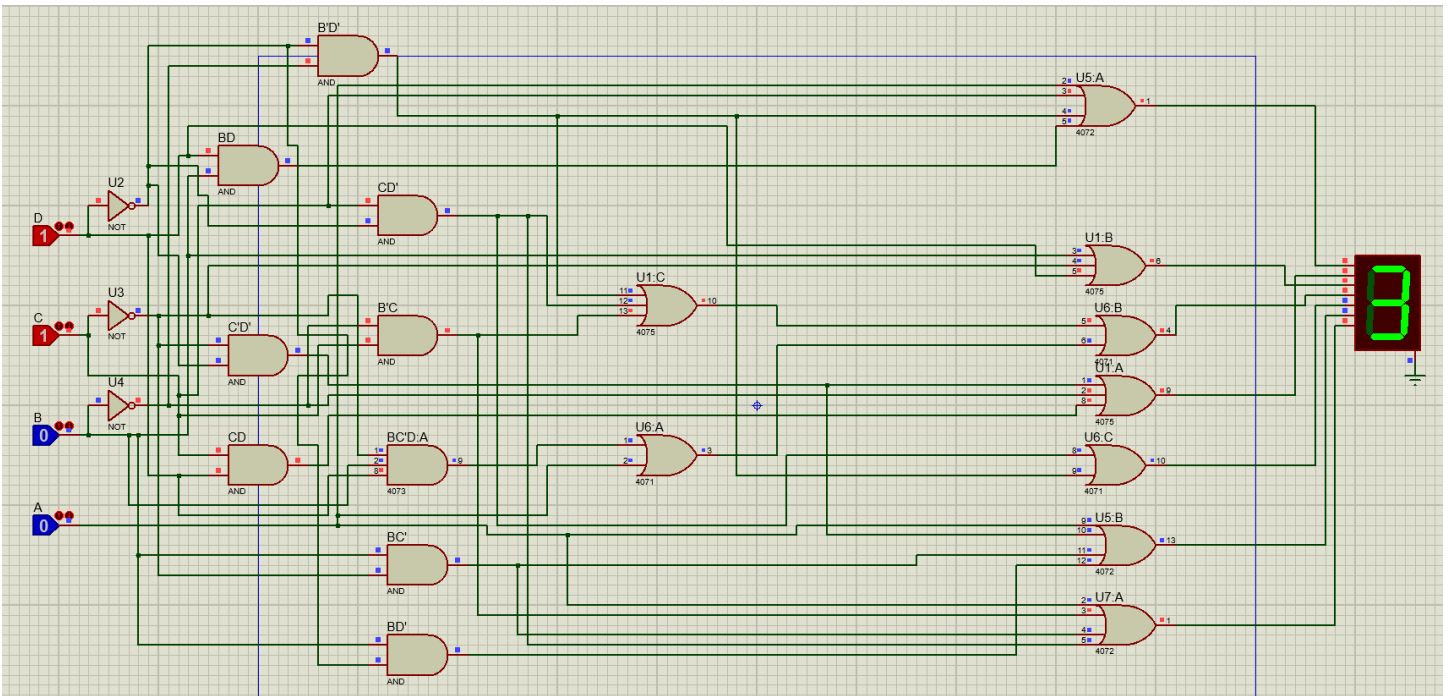
2. Output 1



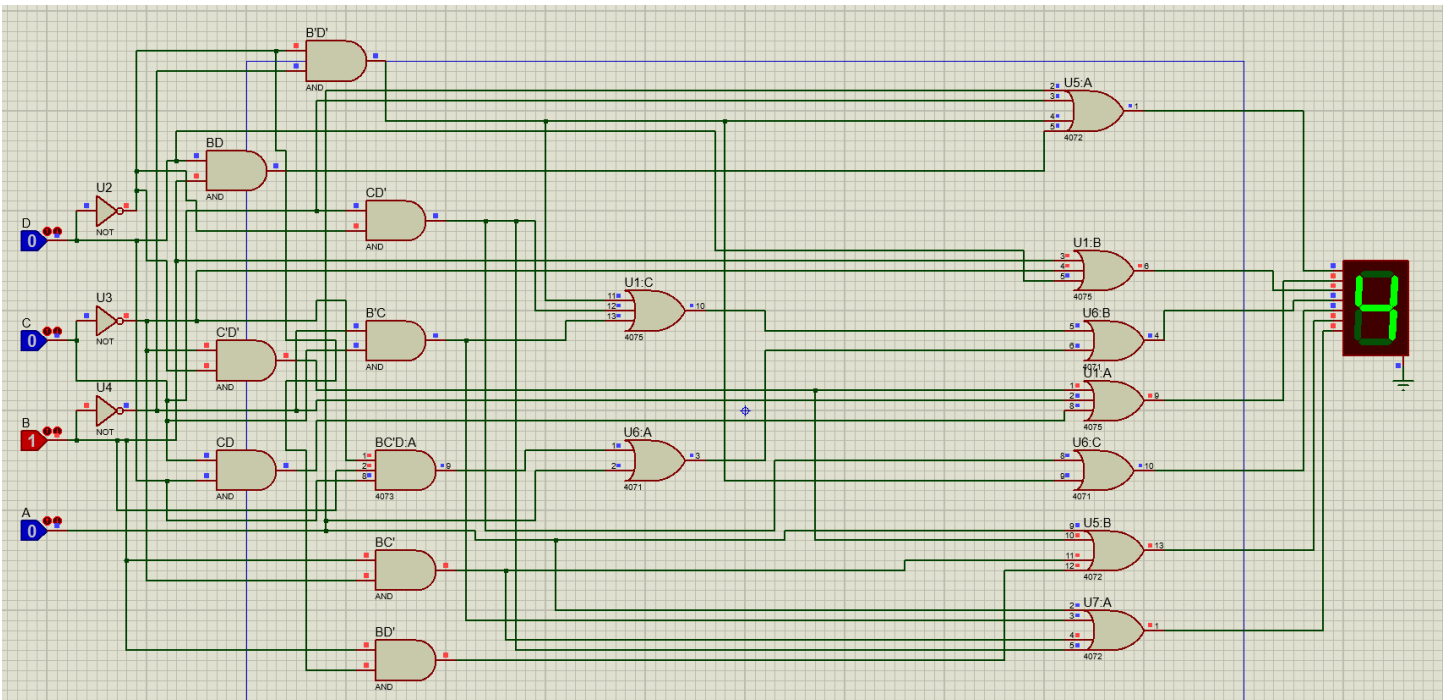
3. Output 2 is displayed:



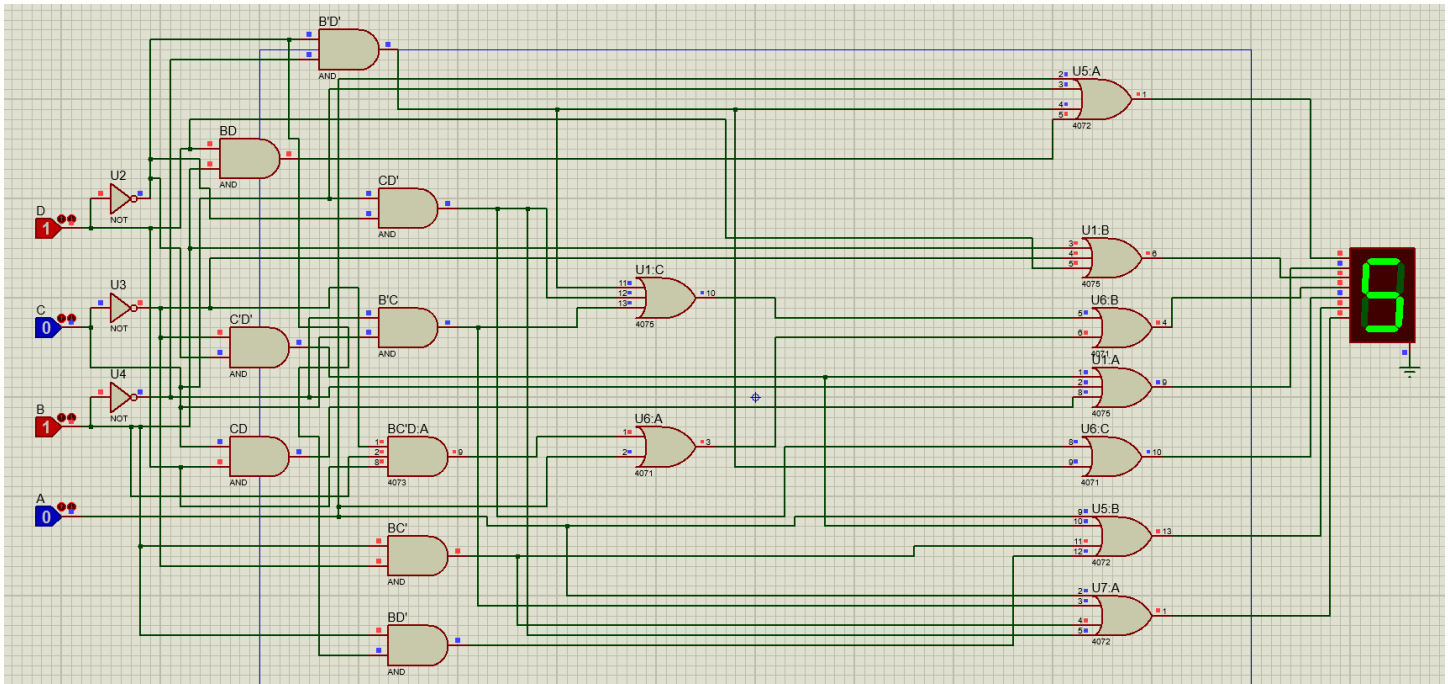
4. Output 3 is displayed:



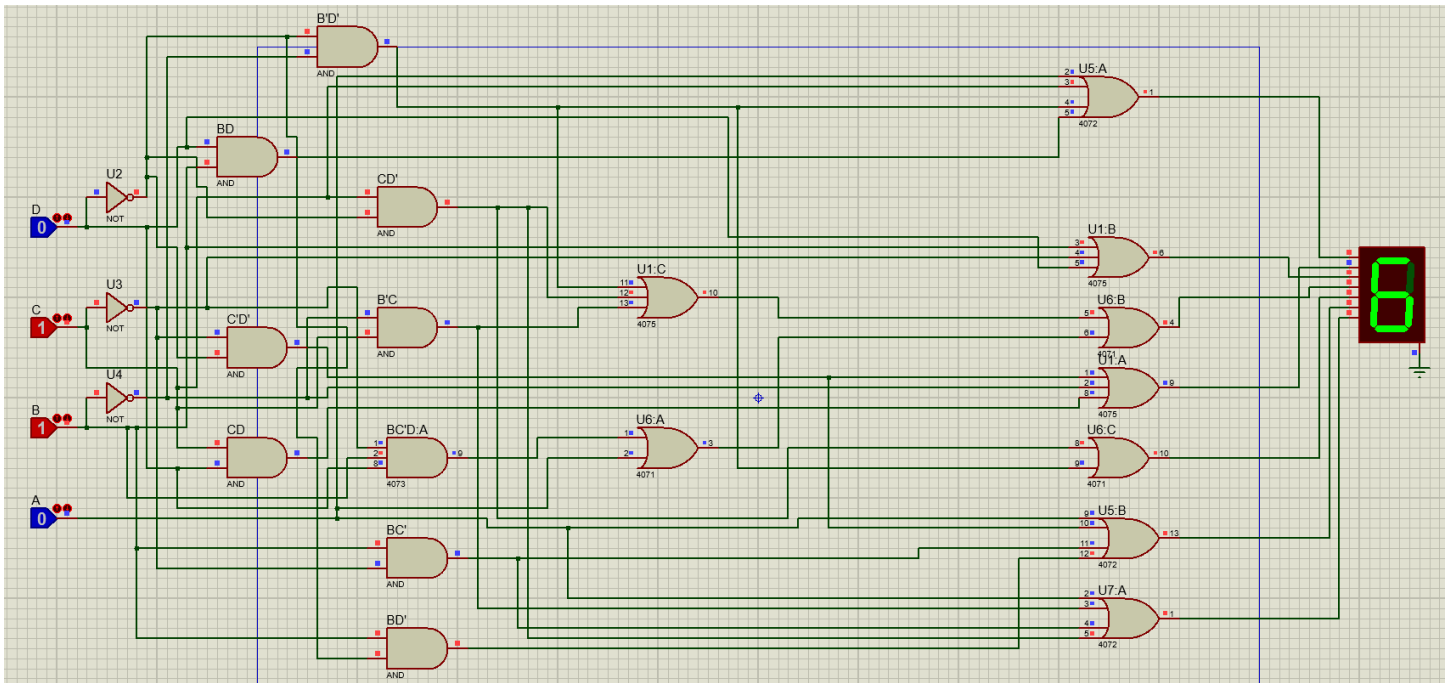
5. Output 4 is displayed:



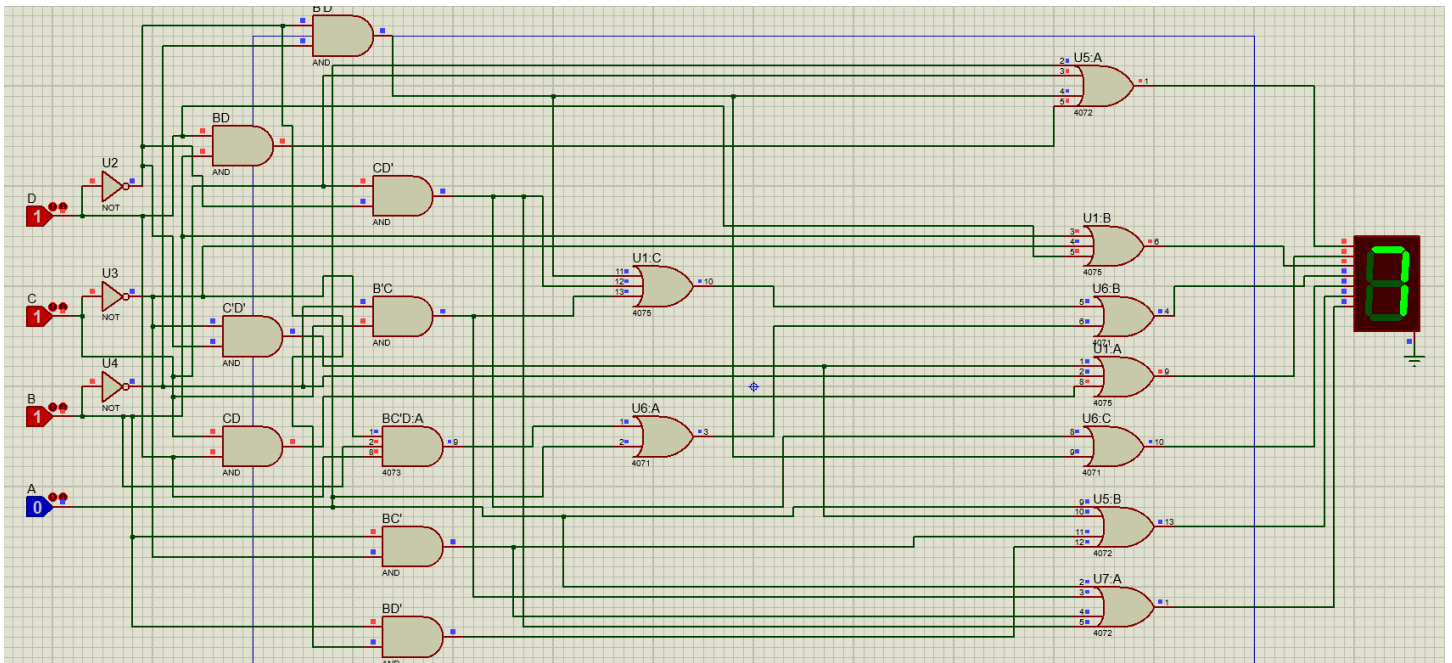
6. Output 5 is displayed:



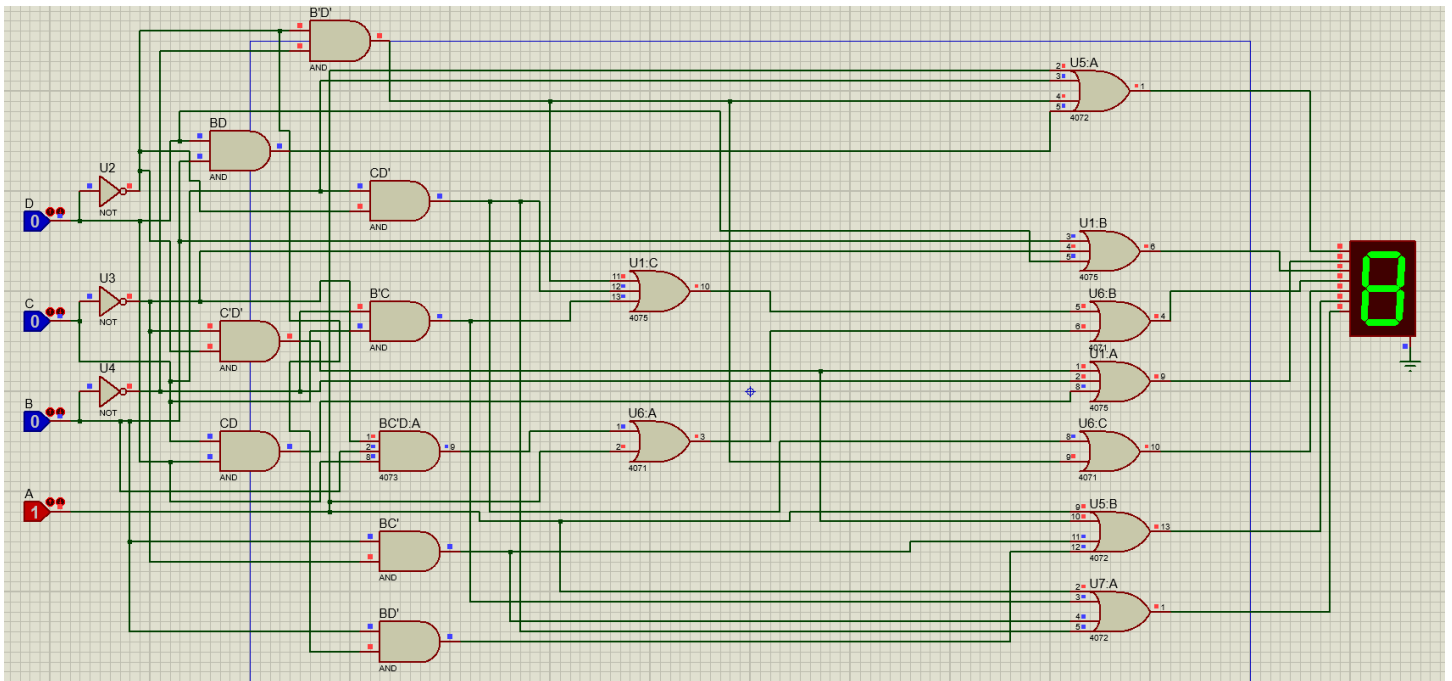
7. Output 6 is displayed:



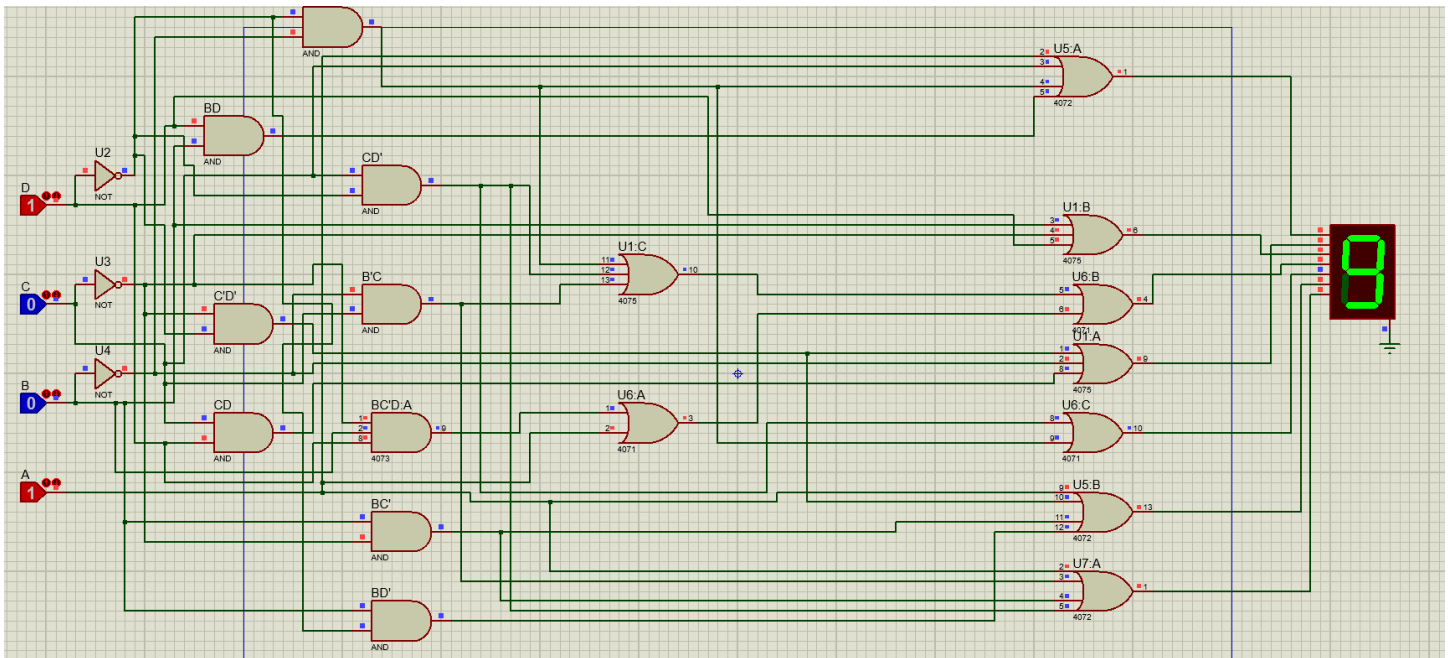
8. Output 7 is displayed:



9. Output 8 is displayed:



10. Output 9 is Displayed:



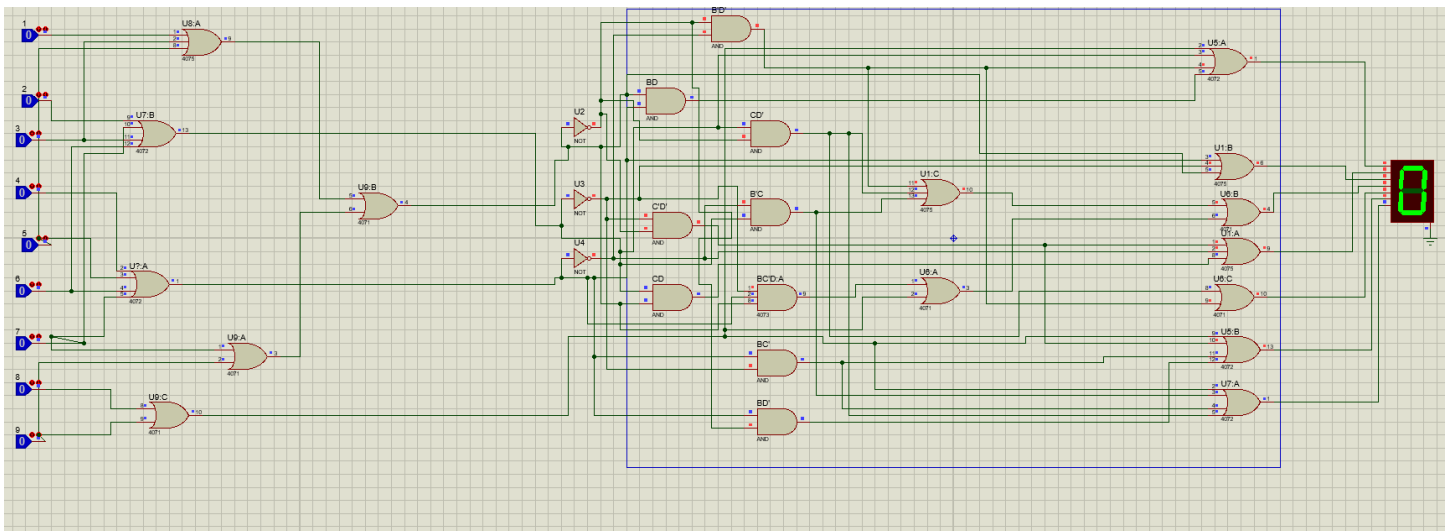
Task 3: Decimal to Seven Segment Encoder (Integration Task):

Procedure:

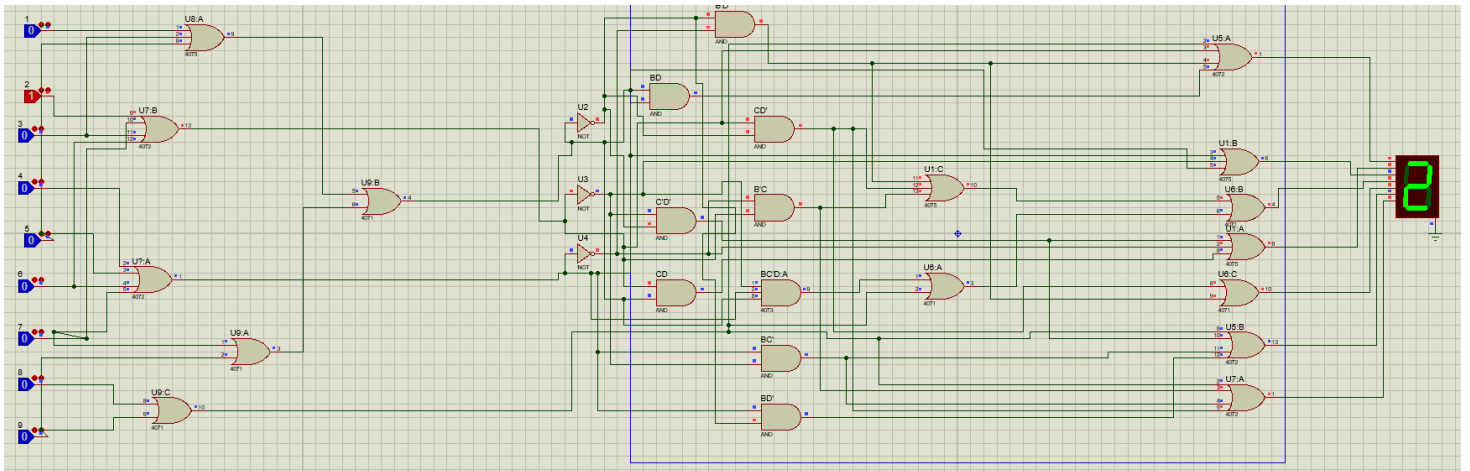
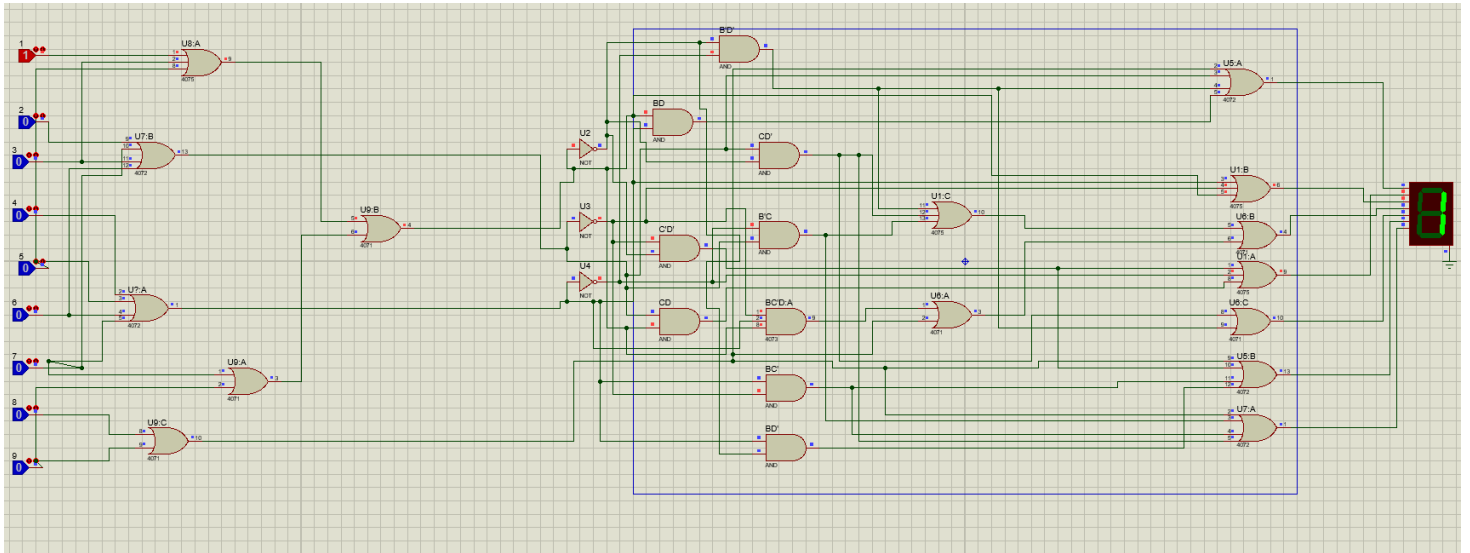
1. Output of the decimal-to-binary encoder was connected directly to the decoder inputs.
2. Decimal buttons (0–9) were pressed one at a time.
3. The correct digit was verified on the 7-segment display.

1. Proteus Simulation Screenshots:

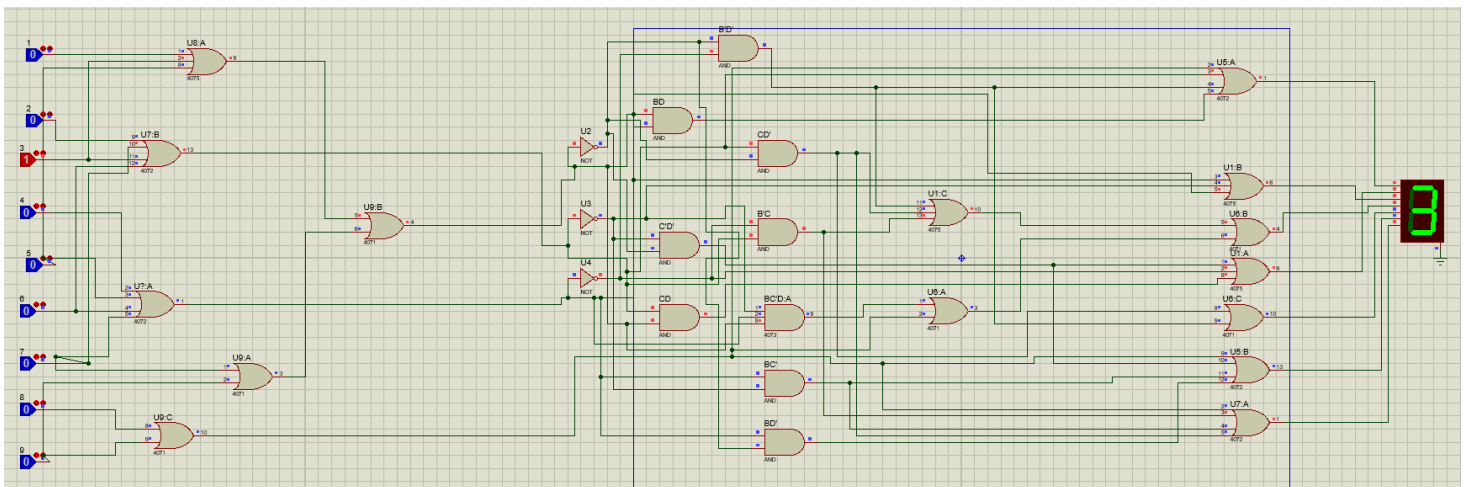
1. When 0 is displayed:



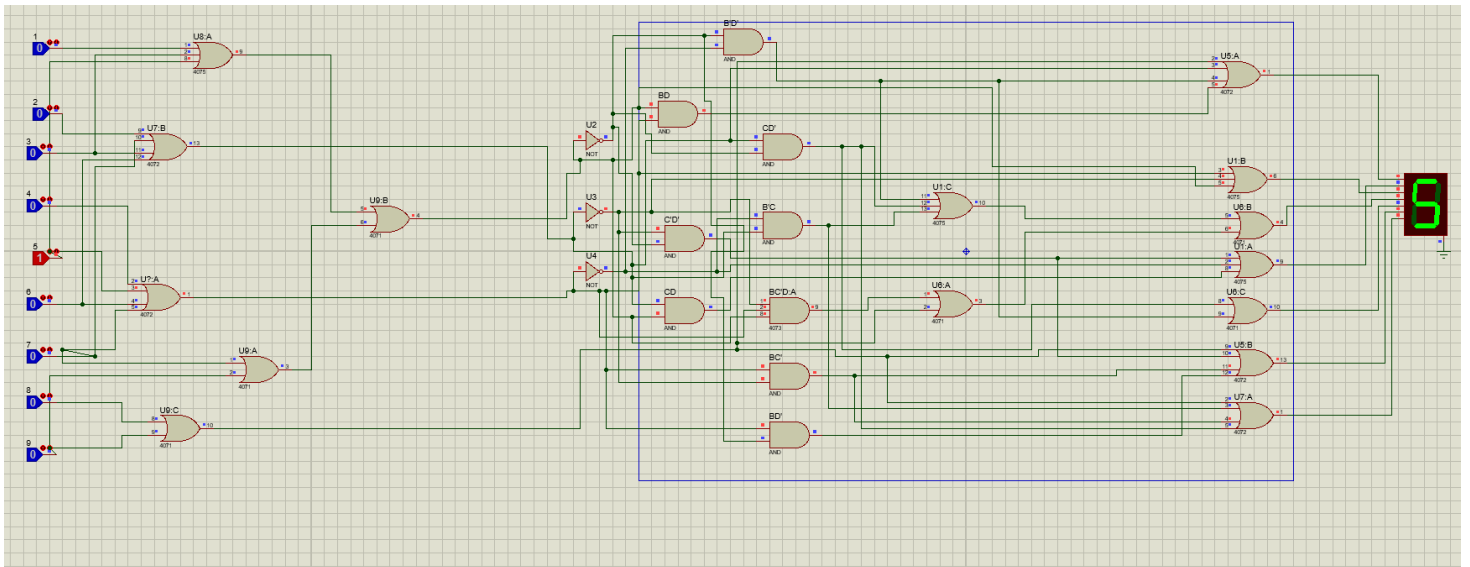
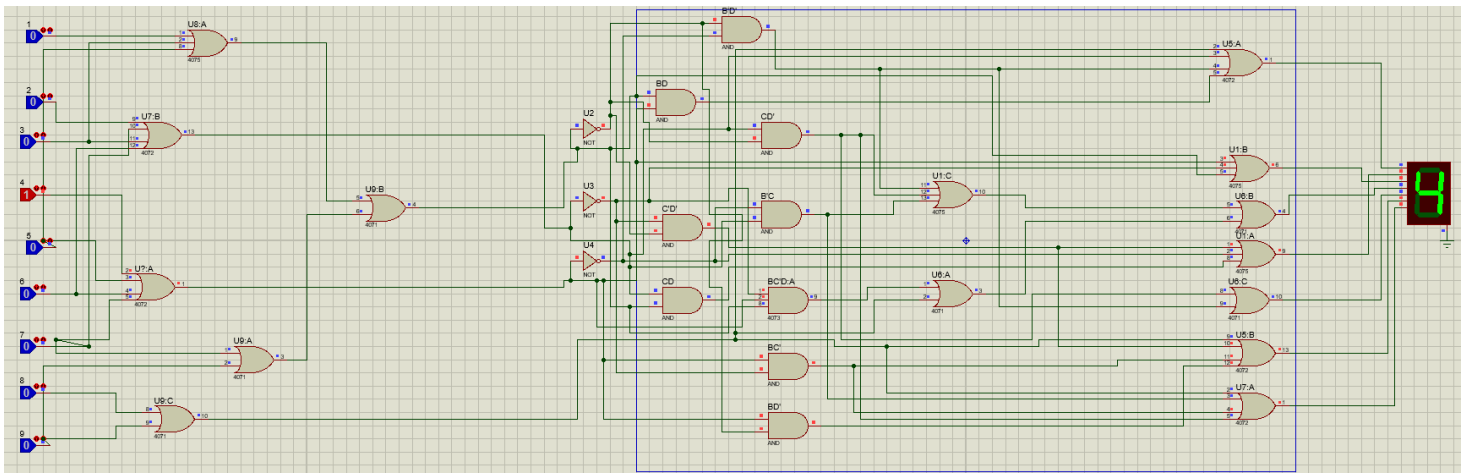
2. 1



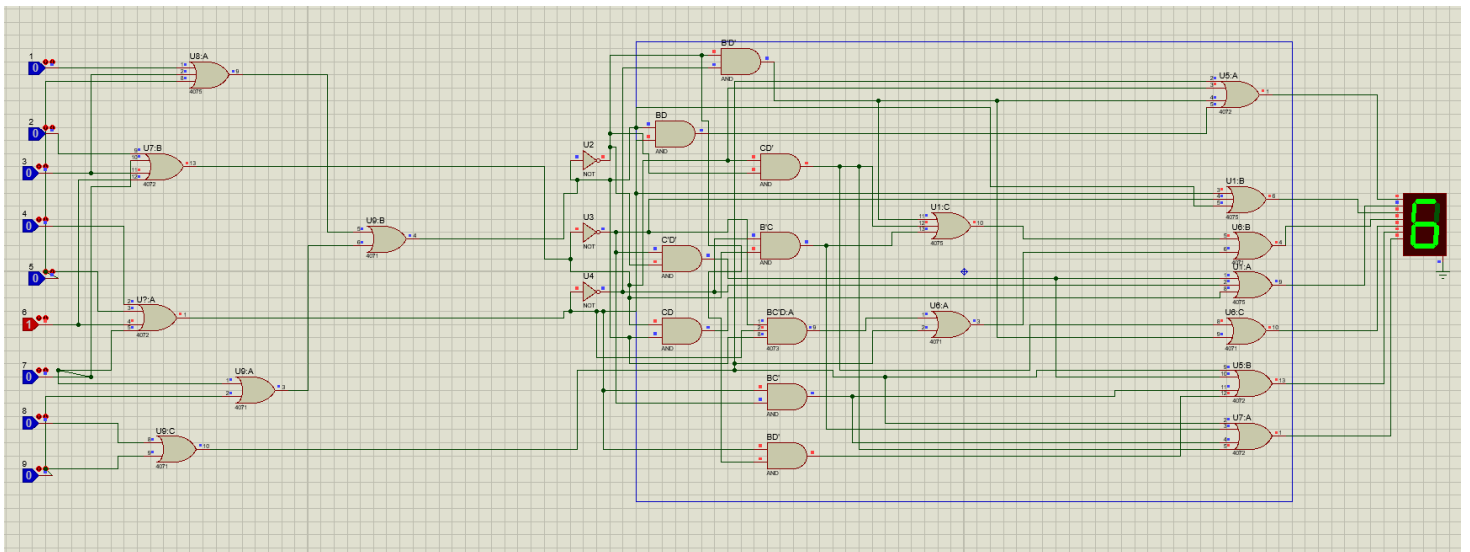
4. When 3 is displayed:



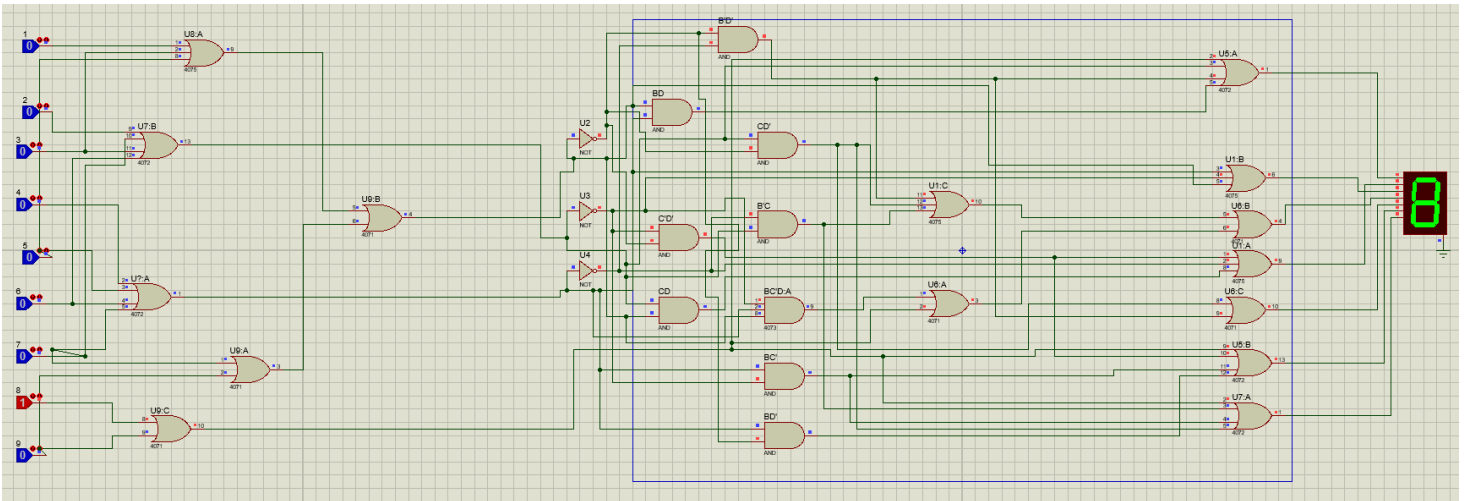
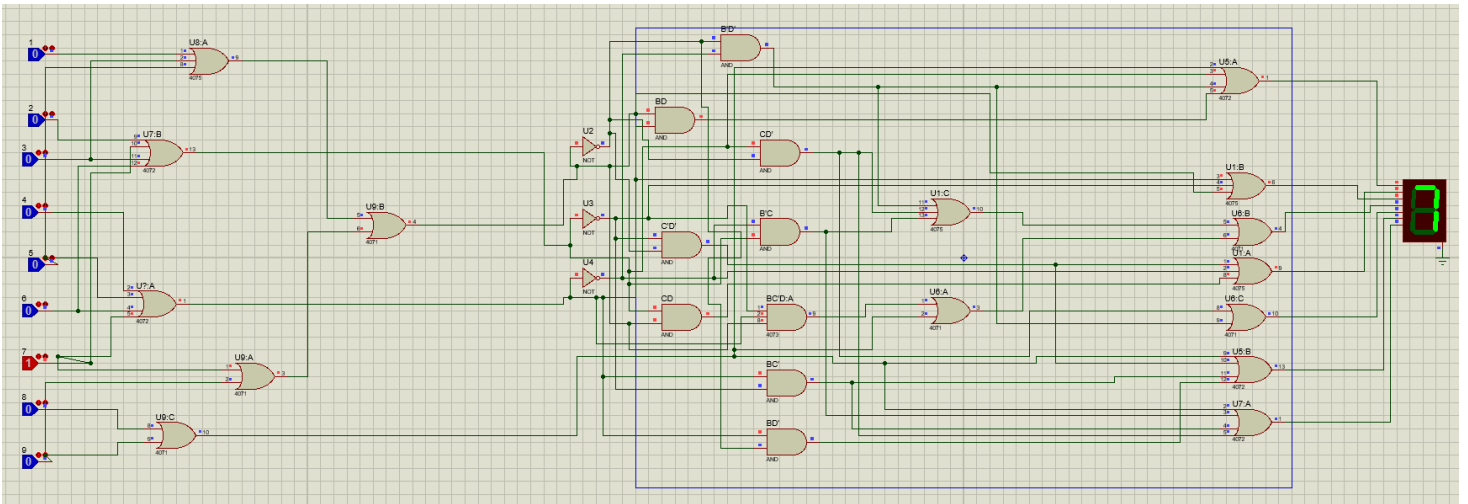
5. 4



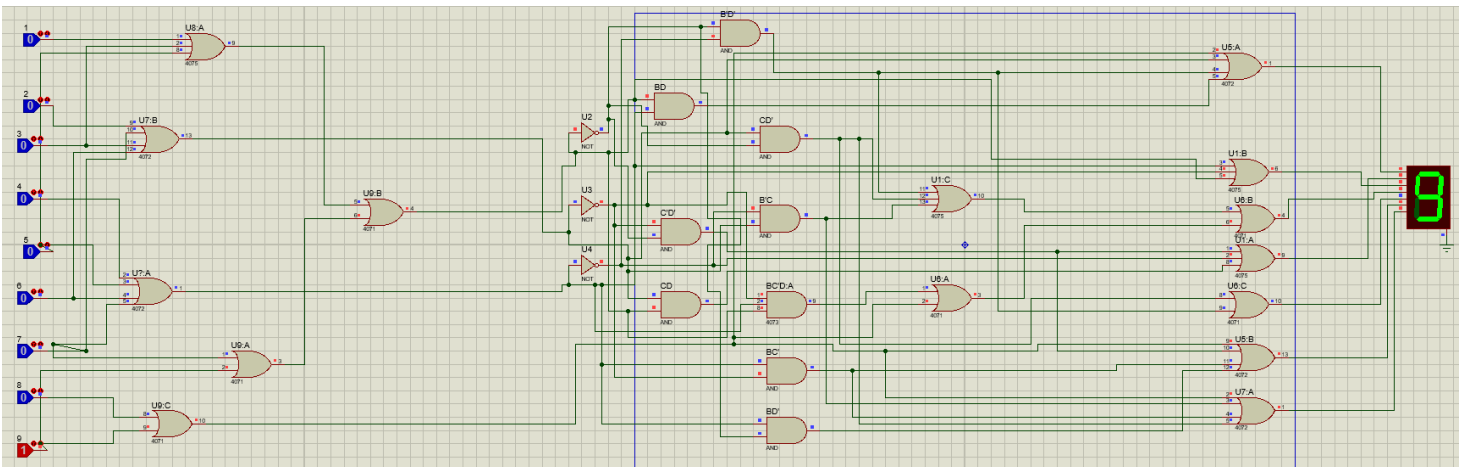
7. When 6 is displayed:



8. 7



10. When 9 is displayed:



Observations and Results

Encoder (Decimal-to-Binary Conversion)

The decimal-to-binary encoder circuit, constructed exclusively with OR gates¹¹¹, demonstrated robust and deterministic functionality:

- **Logic Verification:** The Boolean expressions were confirmed; for example, input **8** or **9** activated the **D0** output (LSB), verifying the logic $D0 = 8 + 9^2$. Similarly, input **7** activated all three higher bits (**D1**, **D2**, **D3**), resulting in the binary code 0111³.
- **Unique Output Encoding:** Each of the ten unique decimal inputs (0–9) reliably activated a single corresponding input line, which in turn generated a unique 4-bit binary code (D3 D2 D1 D0)⁴⁴⁴⁴.
- **Circuit Stability:** The use of pull-down resistors ensured that when no input button was pressed (decimal input 0), the output was stable at the binary code 0000, and no unwanted outputs were observed during switching (no glitches)⁵⁵⁵⁵.

Decoder (Binary-to-Seven-Segment Conversion)

The binary-to-seven-segment decoder, implemented using a combination of AND, OR, and NOT gates, successfully executed the data expansion task:

- **Segment Activation:** The logic circuits for the seven segments (**a** through **g**) correctly interpreted the 4-bit binary input to activate the necessary combination of segments for digits 0-9.
- **Minimized Logic Performance:** The minimized Boolean expressions (e.g., $a = A + C + B \cdot D + B' \cdot D'$) functioned as expected, efficiently driving the 7-segment display for all valid BCD codes (0000 to 1001).
- **Visual Confirmation:** The segmented output clearly and accurately formed the intended decimal digit for every corresponding binary input, validating the decoder's truth table.

Combined System (Decimal-to-Display Integration)

The integrated system seamlessly achieved the end-to-end goal of converting a momentary decimal push-button press into a stable visual output:

- **Complete Signal Path:** Connecting the encoder's 4-bit output (D3, D2, D1, D0) directly to the decoder's 4-bit input (A, B, C, D) created a functional and immediate decimal-to-display system.
- **Real-time Response:** The cascaded system—Encoder → Decoder → Display—operated without perceptible delay, demonstrating the high speed of combinational logic circuits.
- **Practical Validation:** The circuit successfully replicated the core functionality of a digital display system found in devices like calculators and counters, converting a physical input into a meaningful visual output.

Conclusion:

This experiment successfully achieved a comprehensive understanding of how **encoder and decoder** operations function within digital display systems.

- **Task 1 (Encoder):** The **decimal-to-binary encoder** was successfully built using OR gates and verified in Proteus, confirming its ability to generate the correct 4-bit binary codes for decimal digits 0 through 9.
- **Task 2 (Decoder):** The **binary-to-seven-segment decoder** was successfully created using fundamental logic gates, and it correctly activated all necessary segments to form the visual digits.
- **Integration:** The final combined system demonstrated the practical conversion of user decimal input into visual output, verifying how digital systems transform inputs into displays.

The practical application confirmed core concepts of combinational logic, Boolean expressions, and the validity of the designed circuits through both truth table verification and Proteus simulation.

